(13) (i) Given,

$$y = \sin^{-1} x$$

differentiating w.the .r. of x, we get

$$y_1 = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 (\sqrt{1-x^2}) = 1$$

$$\Rightarrow y_1^2 (1-x^2) = 1$$

again, differentiating w.t.r.o.x, we get

$$\Rightarrow 2y_1 y_2 (1-x^2) + y_1^2 (0-2x) = 0$$

$$\Rightarrow (1-x^2) y_2 \cdot 2y_1 - 2xy_1 \cdot 2y_1 = 0$$

$$\Rightarrow (1-x^2) y_2 - xy_1 = 0$$

(11) From (1):

$$(1-x^2)y_2 - xy_1 = 0 \qquad \text{——— (1)}$$

First part, $(1-x^2)y_2$

let,

$u = y_2 \qquad v = (1-x^2)$

$u_1 = y_3 \qquad v_1 = -2x$

$u_2 = y_4 \qquad v_2 = -2$

$u_3 = y_5 \qquad v_3 = 0$

$u_n = y_{n+2}$

by Leibnitz's Theorem,

$$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + \cdots + uv_n$$

$$\{(1-x^2)y_2\}_n = y_{n+2}(1-x^2) + n y_{n+1}(-2x) + \frac{n(n-1)}{2} y_n \cdot (-2) + 0$$

$$= (1-x^2)y_{n+2} - 2nx y_{n+1} - n(n-1)y_n$$

Second part, $-xy_1$

let, $u = y_1 \qquad v = -x$

$u_1 = y_2 \qquad v_1 = -1$

$u_2 = y_3 \qquad v_2 = 0$

$u_n = y_{n+1}$

by Leibnitz's Theorem,

$$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + \cdots + u v_n$$

$$\{-xy_1\}_n = y_{n+1}(-x) + n\, y_n(-1) + 0$$

$$= -xy_{n+1} - ny_n$$

Thus from equation ① ;

$$(1-x^2)\, y_{n+2} - 2nx\, y_{n+1} - n(n-1)y_n - xy_{n+1} - ny_n = 0$$

$$\Rightarrow (1-x^2)\, y_{n+2} - (2n+1)x\, y_{n+1} - (n-1)n\, y_n = 0$$

$$\Rightarrow (1-x^2)\, y_{n+2} - (2n+1)x\, y_{n+1} - n^2 y_n = 0$$

(showed)

(14) (i) Given, $y = \tan^{-1} x$

let's differentiate with the respect to x,

$$y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow y_1(1+x^2) = 1$$

(showed)

(15)

(ii) From equestion (1).

$$(1+x^2)\, y_1 = 1 \quad \text{———} (1)$$

$$\Rightarrow (1+x^2)\, y_1 - 1 = 0$$

~~First part~~

Now differentiate equation (01) with the help of Leibnitz's nth times theorem,

let, $u = y_1$ | $v = 1+x^2$ | $\therefore \frac{d}{dx}(-1)$

$\Rightarrow u_1 = y_2$ | $v_1 = 2x$ | $= 0$

$\Rightarrow u_2 = y_3$ | $v_2 = 2$ | $\therefore {}^nc_1 = n$

$\Rightarrow u_3 = y_4$ | $v_3 = 0$ | $\therefore {}^nc_2 = \frac{n(n-1)}{2}$

$\cdots\cdots$

$\cdots\cdots$

$u_n = y_{n+1}$

by Leibnitz's theorem,

$$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + \cdots\cdots + uv_n$$

$$\{(1+x^2)\, y_1\}_n = y_{n+1}\, (1+x^2) + n\, y_n \cdot 2x + \frac{n(n-1)}{2} \cdot y_{n-1} \cdot 2 + 0$$

$$= (1+x^2)\, y_{n+1} + 2xn\, y_n + n(n-1)\, y_{n-1}$$

(1) ———

$\therefore$ Thus from equation (2)

$$(1+x^2)\, y_{n+1} + 2nx\, y_n + n(n-1)\, y_{n-1} = 0$$

(showed)

(15)

(i) given,

$$\log(y) = \tan^{-1} x$$

differentiate w.r.t. x,

$$\frac{1}{y} \cdot y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2) y_1 = y$$

again differentiate w.r.t. x,

$$y_1 \frac{d}{dx}(1+x^2) + (1+x^2)\frac{d}{dx}(y_1) = \frac{d}{dx}(y)$$

$$\Rightarrow y_1(0+2x) + (1+x^2) \times 1 \cdot y_2 = 1 \cdot y_1$$

$$\Rightarrow 2x y_1 + (1+x^2) y_2 - y_1 = 0$$

$$\Rightarrow (1+x^2) y_2 + (2x-1) y_1 = 0$$

(showed.)

(ii) From question (i)

$$(1+x^2) y_2 + (2x-1) y_1 = 0 \quad \text{——— (1)}$$

Now differentiate equation (1) "nth times" with the help of Leibnitz theorem.

First Part. $(1+x^2)y_2$

let,

$u = y_2$	$v = 1+x^2$
$u_1 = y_3$	$v_1 = 2x$
$u_2 = y_4$	$v_2 = 2$
$u_3 = y_5$	$v_3 = 0$
$\cdots \quad \cdots$	
$u_n = y_{n+2}$	

$\therefore$ by Leibnitz's theorem,

$$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + \cdots\cdots + u v_n$$

$$\therefore \{(1+x^2)y_2\}_n = y_{n+2}(1+x^2) + n \cdot y_{n+1} \cdot 2x + \frac{n(n-1)}{2} \cdot y_n \cdot 2 + 0$$

$$= (1+x^2)y_{n+2} + 2nx\, y_{n+1} + n(n-1)\, y_n$$

Second part, $(2x+1)y_1$

let,

$u = y_1$	$v = (2x+1)$
$u_1 = y_2$	$v_1 = 2$
$u_2 = y_3$	$v_2 = 0$
$\cdots \quad \cdots$	
$u_n = y_{n+1}$	

by leibnitz's theorem,

$$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + \cdots + uv_n$$

$$\{(2x-1)y_1\}_n = y_{n+1}(2x-1) + n \cdot y_n \cdot 2 + 0$$

$$= (2x-1)y_{n+1} + 2n y_n$$

Thus from equation (1)

$$(1+x^2)y_{n+2} + 2nx\, y_{n+1} + n(n-1)y_n + (2x-1)y_{n+1} + 2ny_n = 0$$

$$\Rightarrow (1+x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n+1)y_n = 0$$

(Showed)

(16) (i) Given, $y = (\sin^{-1} x)^2$

Differentiating w.r.t x we get,

$$y_1 = 2\sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1(\sqrt{1-x^2}) = 2\sin^{-1} x$$

$$\Rightarrow y_1^2(1-x^2) = (2\sin^{-1} x)^2$$

$$\Rightarrow (1-x^2)y_1^2 = 4y$$

Again differentiating w.r.t. x, we get,

$$(1-x^2)\cdot 2y_1 y_2 + y_1^2(0-2x) = 4\cdot 1\cdot y_1$$

$$\Rightarrow (1-x^2)y_2 \cdot 2y_1 - xy_1\cdot 2y_1 - 2\cdot 2y_1 = 0$$

$$\Rightarrow (1-x^2)y_2 - xy_1 - 2 = 0$$

Showed.

(ii) From eqns (1),

$$(1-x^2)y_2 - xy_1 - 2 = 0 \quad \text{——— (1)}$$

Now differentiating equation (1) nth times with the help of Leibnitz's theorem,

First part, $(1-x^2)y_2$

let, $u = y_2$	$v = 1-x^2$	Here,
$u_1 = y_3$	$v_1 = -2x$	$^nC_1 = n$
$u_2 = y_4$	$v_2 = -2$	$^nC_2 = \frac{n(n-1)}{2}$
$u_3 = y_5$	$v_3 = 0$	
... ...		
$u_n = y_{n+2}$		

by Leibnitz's theorem,

$$(uv)_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \cdots + uv_n$$

$$\{(1-x^2)y_2\}_n = y_{n+2}(1-x^2) + n\, y_{n+1}(-2x) + \frac{n(n-1)}{2} y_n (-2) + 0$$

$\therefore (1-x^2) y_{n+2} - 2nx\, y_{n+1} - n(n-1) y_n$

Second part, $-xy_1$

let, $u = y_1$ | $v = -x$

$u_1 = y_2$ | $v_1 = -1$

$u_2 = y_3$ | $v_2 = 0$

... ...

... ...

$u_n = y_{n+1}$

by Leibnitz's Theorem,

$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + \cdots + uv_n$

$\{-xy_1\}_n = y_{n+1} \cdot (-x) + n \cdot y_n \cdot (-1) + 0$

$= -x y_{n+1} - n y_n$

Third part, -2

$\Rightarrow \frac{d}{dx}(-2) = 0$

Thus, from equation (1),

$(1-x^2) y_{n+2} - 2nx\, y_{n+1} - n(n-1) y_n - x y_{n+1} - n y_n - 0 = 0$

$\Rightarrow (1-x^2) y_{n+2} - (2n+1) x\, y_{n+1} - n(n-1+1) y_n = 0$

$\Rightarrow (1-x^2) y_{n+2} - (2n+1) x\, y_{n+1} - n^2 y_n = 0$

(Showed)